Bitcoin Market Sentiment vs. Trader Performance Analysis

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1. Introduction

Financial markets are strongly influenced by investor emotions. In the cryptocurrency market, fear and greed often affect trading decisions more than traditional financial indicators. During periods of fear, traders may panic and sell their assets, while during periods of greed, they are more likely to take larger risks in anticipation of higher returns.

The purpose of this project is to understand how market sentiment influences trader performance. Using historical trading data from Hyper liquid together with the Bitcoin Fear & Greed Index, this analysis investigates whether traders perform differently under various market conditions. Rather than focusing only on profits, the study also examines trading activity, trade size, winning percentage, and asset-wise performance.

The overall objective is to identify meaningful patterns that could help traders make better decisions by considering market sentiment alongside historical trading behaviour.

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2. Objective

The primary objective of this project is to analyse the relationship between Bitcoin market sentiment and trader performance.

The analysis focuses on answering the following questions:

- Does market sentiment influence the number of trades executed?
- Which sentiment category generates the highest profits?
- Does the average profit per trade change across different market conditions?
- Which traders consistently perform well?
- Which cryptocurrencies generate the highest returns?
- How does trade size vary under different market sentiments?
- Can market sentiment be used as an additional factor while designing trading strategies?

By answering these questions, the project aims to provide practical insights that may help improve trading decisions.

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3. Dataset Description

Two datasets were provided for this assignment.

Historical Trading Dataset

This dataset contains detailed trading information collected from Hyperliquid. Each record represents an individual trade executed by a trader.

Important attributes include:

- Account ID
- Coin
- Execution Price
- Trade Size
- Trade Value (USD)
- Buy/Sell Side
- Timestamp
- Closed Profit and Loss
- Trading Fee

This dataset forms the foundation for analysing trader performance.

Bitcoin Fear & Greed Index

The second dataset contains daily Bitcoin market sentiment.

Each day is classified into one of the following categories:

- Fear
- Greed
- Neutral
- Extreme Greed

This dataset was merged with the trading dataset using the trading date so that every trade could be analysed within its corresponding market sentiment.

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4. Data Preparation

Before starting the analysis, the datasets were cleaned and prepared.

The historical trading dataset contained timestamps in Unix format, which were converted into readable datetime values. Similarly, the sentiment dataset dates were converted into a common datetime format.

A new Date column was created in both datasets, allowing them to be merged successfully. After merging, each trade record included its corresponding market sentiment.

Additional preprocessing included:

- Verifying column data types
- Checking for duplicate records
- Inspecting missing values
- Creating new analytical variables such as the Win column, where profitable trades were marked as successful trades

Proper data preparation ensured that the analysis was based on clean and reliable data.

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5. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the relationship between market sentiment and trader performance.

Several visualizations were created to examine different aspects of the trading data.

Trade Distribution

The first visualization shows the number of trades executed under different market sentiment conditions.

The results indicate that Fear periods recorded the highest trading activity. This suggests that traders tend to participate more actively during uncertain market conditions, possibly because increased price volatility creates more trading opportunities.

In comparison, Greed, Neutral, and Extreme Greed periods recorded fewer trades.

(Insert Trade Distribution Chart)

Total Profit Analysis

The total Closed Profit and Loss was calculated separately for each market sentiment category.

The analysis shows that Fear generated the highest overall trading profit. Since more trades were executed during Fear, the cumulative profit naturally became larger.

Greed contributed the second-highest total profit, while Neutral generated the lowest overall returns.

Although total profit is useful for measuring overall performance, it should be interpreted together with trading volume.

(Insert Total Profit Chart)

Average Profit Per Trade

To remove the effect of trading frequency, the average Closed Profit per trade was calculated.

Interestingly, Greed produced the highest average profit per trade even though it had fewer trades than Fear.

This observation suggests that traders may execute fewer but more carefully selected trades during optimistic market conditions.

The result highlights that profitability cannot be judged solely by the total number of trades.

(Insert Average Profit Chart)

Win Rate Analysis

A trade was considered successful whenever its Closed Profit and Loss was greater than zero.

The analysis shows that Extreme Greed achieved the highest winning percentage among all market conditions.

Fear recorded a moderate win rate despite having the highest number of trades.

This indicates that higher trading activity does not necessarily guarantee a higher probability of success.

(Insert Win Rate Chart)

Top Performing Traders

Trader-wise profitability was calculated by summing the Closed Profit and Loss for each account.

The analysis clearly shows that a relatively small number of trader accounts generated a significant portion of the total profits.

This concentration of profits suggests that experienced traders are able to consistently outperform the average participant.

Such traders may possess better risk management techniques, stronger market knowledge, or more disciplined trading strategies.

(Insert Top Traders Chart)

Coin-wise Performance

The profitability of different cryptocurrencies was also analysed.

Several alternative cryptocurrencies generated higher cumulative profits than Bitcoin.

Assets such as HYPE, SOL, ETH, and @107 appeared among the highest-performing coins.

This observation indicates that profitable trading opportunities are distributed across multiple cryptocurrencies rather than being limited to Bitcoin alone.

(Insert Coin-wise Profit Chart)

Average Trade Size

The average trade value (USD) was calculated for each sentiment category.

Extreme Greed recorded the largest average trade size, indicating that traders tend to invest larger amounts when market confidence is high.

Fear also showed relatively large trading positions, reflecting increased activity during volatile market conditions.

(Insert Trade Size Chart)

Correlation Analysis

A correlation heatmap was generated to understand relationships between numerical variables.

The strongest positive relationship was observed between Trade Size (USD) and Trading Fee, which is expected because larger trades naturally incur higher fees.

Profitability, however, showed only weak correlations with the available numerical variables.

This suggests that trading success depends on multiple factors, including timing, market conditions, and trading strategy.

(Insert Heatmap)

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6. Key Findings

The analysis produced several interesting findings.

- Trading activity was highest during Fear periods.
- Fear generated the highest total profit because of its larger trading volume.
- Greed achieved the highest average profit per trade.
- Extreme Greed produced the highest win rate.
- A small number of traders generated a substantial share of the overall profits.
- Alternative cryptocurrencies outperformed Bitcoin in cumulative profitability.
- Larger trades resulted in higher trading fees.
- Market sentiment clearly influences trader behaviour as well as trading outcomes.

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7. Conclusion

This project successfully explored the relationship between Bitcoin market sentiment and trader performance by combining historical trading records with the Bitcoin Fear & Greed Index.

The findings demonstrate that market sentiment influences not only how frequently traders participate but also how effectively they perform. Fear periods were associated with increased trading activity and the highest cumulative profits, while Greed periods generated the highest average returns per trade. Extreme Greed exhibited the highest winning percentage despite having fewer trades.

Overall, this analysis shows that market sentiment can serve as a valuable indicator when evaluating trading behavior. While sentiment alone cannot predict profitability, combining it with technical analysis, risk management, and historical trading patterns can help traders make more informed decisions.

8. Future Improvements

This project can be extended in several ways to provide deeper insights.

Future work could include:

- Developing machine learning models to predict profitable trades.
- Performing trader segmentation based on trading behavior.
- Calculating advanced performance metrics such as the Sharpe Ratio and maximum drawdown.
- Building an interactive dashboard using Streamlit or Power BI.
- Incorporating additional market indicators such as trading volume and volatility.

These enhancements would provide a more comprehensive understanding of trader behavior and improve the practical value of the analysis.